TR-M318-Ram-Ramsey

A Classification of Finite Euclidean Ramsey Configurations: Exact Tensor Certificates, Ram Moment Stencils, and a Step-Level Proof Audit

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Abstract

We analyze the September 23, 2026 manuscript of OpenAI at the exact snapshot identified in Section 1. Its classification concerns arbitrary finite colorings, congruence at the original scale, and an ambient dimension allowed to depend on the number of colors. We reconstruct the coordinate-field descent, tensor-separation coloring, Laurent-ideal stencil, signed Gaussian redistribution, noncommutative endpoint correction, and finite color-synchronization arguments. The useful Ram contribution is an exact signed moment representation with an explicit remainder and constraint-preserving rationalization, not a renamed transform or a replacement for the combinatorial forcing theorem.

Independent local extensions include an explicit separability-idempotent certificate for algebraic spherical configurations, a finite rational linear-system alternative over number fields, a quantitative rounding budget, and a smaller universal obstruction-color bound. Full proofs are given. The at-most-five circle-point interpolation and the generic nine-point obstruction are derived, together with the source's twelve-point derivation obstruction. An independently checked derivation-to-tensor bridge recovers the seven-point construction of Pálvölgyi's Appendix A.3 without importing its real-coefficient coloring theorem. A rational cyclic kite separates tensor feasibility from box/cut-cone feasibility with an exact positive distance to the cut cone. The source proof is audited at the mathematical-argument level; this is not a formal verification, a priority determination, or validation of the whole manuscript collection. All unexecuted and external gates are identified. L5 applicability is not global validation.

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1 Scope, provenance, and exact quantifiers

The target is catalogue row 318, family 172, *A classification of finite Euclidean Ramsey configurations*, by OpenAI [1]. Both the local catalogue and source inventory identify commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The extracted source is not a Git worktree. Provenance is therefore checked against the snapshot and per-file SHA-256 inventory, not an invented local Git HEAD. We read the complete main file, bibliography, all seven mathematical sections, and the synchronization figure. The actual private `TR-item-229-astra.tex` was read, including its appended constructive cut coordinates and exact path-repair theorem. Its September 7 historical claims of openness are not silently promoted to current literature status. Precise paths and reading depths appear in Section 12.

Definition 1.1 (The property being classified)**.** A nonempty finite configuration $A \subset \mathbb{R}^d$ is Ramsey when

$$\begin{aligned} &\forall r \geq 2\ \exists D = D(A,r) \geq 1\ \forall c : \mathbb{R}^D \to [r]\ \exists \{b_1, \dots, b_s\} : \\ &c(b_1) = \cdots = c(b_s), \qquad \|b_i - b_j\| = \|a_i - a_j\| \quad \text{for all } i, j. \end{aligned} \tag{1.1}$$

Here $A = \{a_1, \dots, a_s\}$ consists of distinct points. No measurability, continuity, density, or definability hypothesis is imposed on c.

The scale in (1.1) is one. Allowing an unspecified homothety inside the last existential quantifier is a different problem. Nevertheless the property of a *fixed* configuration is invariant under a fixed similarity: pull a coloring back by that similarity. This does not permit selecting a scale after observing an arbitrary coloring. Singletons are immediate; otherwise we work in the affine span, so $d \geq 1$. If $D < d$ no congruent copy exists; countercolorings there are automatic.

What is and is not claimed. Sections 2–7 reconstruct the source argument and check its displayed proof obligations. The classification is used with the explicitly stated classical Hales–Jewett and Noetherian completion-flatness inputs. These inputs are not reproved from foundations. ZFC basis choice and compactness are also used, as in the source. The finite checker verifies selected exact identities, not those infinite arguments. A complete machine-checked identification with any advertised formal theorem, effective bounds for the forcing dimension, and recovery of exact coordinate fields from numerical data remain unverified gates. Positive consequences are conditional on the full sufficiency chain, not on the success of our finite tests. Negative examples below need only the independently reconstructed necessity argument.

2 Tensor criterion and its necessary direction

Put $p_i = (1, a_i)^T$, $F = \mathbb{Q}((a_i)_\alpha)$, and $B = F \otimes_{\mathbb{Q}} F$. Multiplication $m : B \to F$ sends $x \otimes y$ to xy. The source criterion is

$$\begin{aligned} E_i(P) &= (p_i \otimes 1)^T P (1 \otimes p_i) = 0 \quad (1 \leq i \leq s), \\ m(P_{\alpha\beta}) &= \delta_{\alpha\beta} \quad (1 \leq \alpha, \beta \leq d). \end{aligned} \tag{2.1}$$

There is no separate restriction on the constant row or column. The existence of $P \in \mathrm{Mat}_{d+1}(B)$, not positivity of P, is the assertion. The source claims that this is equivalent to (1.1).

2.1 The field descent, including the symmetry convention

Let $W = \mathbb{R} \times \mathbb{R}^d$ as a *rational* vector space and $e_i(v,u) = v + a_i \cdot u$. Let $\mathcal{T}(W)$ be the flip-invariant subspace of $W \otimes_{\mathbb{Q}} W$. Define

$$G((v,u) \otimes (v',u')) = u(u')^T.$$

On symmetric tensors this is a symmetric matrix. The affine-row version is

$$(e_i \otimes e_i)T = 0 \text{ in } \mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R}, \qquad G(T) = I_d. \tag{2.2}$$

Proposition 2.1 (Exact descent; source `alg:equivalence`)**.** *Conditions* (2.1) *and* (2.2) *are equivalent.*

Proof. Arrays over $\mathcal{B} = \mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R}$ identify with $W \otimes_{\mathbb{Q}} W$. If $\tau(x \otimes y) = y \otimes x$, tensor flip is $Q^*_{\alpha\beta} = \tau(Q_{\beta\alpha})$. Then $E_i(Q^*) = \tau(E_i(Q))$, and $m(Q^*) = m(Q)^T$. Thus $(Q + Q^*)/2$ retains all evaluation zeros and the identity spatial block. Inclusion $F \hookrightarrow \mathbb{R}$ induces an injective map on rational tensor products. This proves the forward implication.

For the converse choose an F-basis (r_λ) of $\mathbb{R}$. There is a free B-module decomposition

$$\mathcal{B} = \bigoplus_{\lambda,\mu} B(r_\lambda \otimes r_\mu).$$

Write the finitely supported expansion of Q as $Q = \sum (r_\lambda \otimes r_\mu) P^{\lambda\mu}$. The B-linearity of E_i implies $E_i(P^{\lambda\mu}) = 0$ separately. Let $C^{\lambda\mu}$ be their multiplied spatial blocks. Then $I_d = \sum r_\lambda r_\mu C^{\lambda\mu}$. This is a finite consistent linear system over F with a real solution. Row reduction over F gives coefficients $c_{\lambda\mu} \in F$ with $I_d = \sum c_{\lambda\mu} C^{\lambda\mu}$. Consequently $P = \sum (c_{\lambda\mu} \otimes 1) P^{\lambda\mu}$ works. The $C^{\lambda\mu}$ need not be symmetric; descent takes place in the full matrix space before the optional symmetrization. □

Proposition 2.2 (Congruence and the ordinary sphere equation)**.** *The tensor condition is invariant under congruence, even if the coordinate field changes. It implies sphericity, but multiplication alone loses data.*

Proof. For $a_i' = Qa_i + t$, $Q \in O(d)$, set $H(v,u) = (v - t \cdot Qu, Qu)$. This is an invertible rational-linear map, $e_i' H = e_i$, and $G((H \otimes H)T) = QG(T)Q^T$. For $J = m(P)$, multiplying (2.1) gives

$$\|a_i\|^2 + \ell \cdot a_i + c = 0, \quad \ell_\alpha = J_{0\alpha} + J_{\alpha 0}, \quad c = J_{00}.$$

Completing the square gives the common center $-\ell/2$. The radius is positive for at least two distinct points. For a transcendental t, $t \otimes 1 - 1 \otimes t$ is a nonzero rational tensor annihilated by m. Extend $\{1, t\}$ to a rational basis and apply the two coordinate functionals to prove nonzeroness. Thus ordinary quadratic equalities cannot replace tensor equalities. □

The same warning already occurs over a number field. For $F = \mathbb{Q}(\sqrt{2})$, let σ be its nontrivial automorphism. The map $x \otimes y \mapsto x\sigma(y)$ sends $\sqrt{2} \otimes 1 - 1 \otimes \sqrt{2}$ to $2\sqrt{2}$, whereas multiplication sends it to zero. Thus the loss is not restricted to transcendental coordinates.

2.2 Separation and an explicit dimension-independent coloring

Theorem 2.3 (Necessity and a local color-bound refinement)**.** *If* (2.2) *has no solution, then in every dimension there is an avoiding coloring using at most* $(2s-2)^{s-1}$ *colors. In particular a Ramsey configuration satisfies* (2.1)*.*

Proof. Regard $\mathcal{Y} = (\mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R})^s \oplus \mathrm{Sym}_d(\mathbb{R})$ over $\mathbb{Q}$ and let $\Phi(T) = ((e_i \otimes e_i)T)_i \oplus G(T)$. The point $y_0 = (0, \dots, 0, I_d)$ is outside its image. Choose a rational basis of the quotient containing the class of y_0; the coordinate functional ℓ satisfies $\ell\Phi = 0$, $\ell(y_0) = 1$. Write its components as ℓ_i and L, and set $h_i(x) = \ell_i(x \otimes x)$. On a pure square,

$$\sum_i h_i(v + a_i \cdot u) + L(uu^T) = 0, \qquad L(I_d) = 1. \tag{2.3}$$

Taking $u = 0$ gives $\sum_i h_i(x) = 0$. For $z \in \mathbb{R}^D$ put $H_i(z) = \sum_{j=1}^D h_i(z_j)$. Every congruent copy has the form $b_i = t + Qa_i$, $Q^TQ = I_d$: polarization gives equality of the Gram matrices of point differences, and hence a well-defined isometry on their span. Apply (2.3) to each row u_j^T of Q. Finite additivity yields

$$\sum_i H_i(b_i) = -1, \qquad \sum_i H_i(z) = 0.$$

Partition $[0, 2)$ into $K = 2(s-1)$ half-open intervals. Color z by the intervals containing $H_2(z), \dots, H_s(z)$ modulo 2. If a congruent copy were monochromatic, then for $i \geq 2$

$$H_i(b_i) - H_i(b_1) = 2n_i + \epsilon_i, \quad |\epsilon_i| < 2/K = 1/(s-1).$$

The difference for $i = 1$ is zero. Summing gives $-1 = 2\sum_{i=2}^s n_i + \sum_{i=2}^s \epsilon_i$, impossible because the last sum has absolute value strictly less than one. This proves the bound. The source uses $(2s+1)^s$ colors; omission of the redundant first functional and retention of the strict half-open-interval inequality give the refinement. No regularity of the functionals is needed. □

Remark 2.4 (Why closed-cone separation is not a substitute). The proof separates a point from a rational-linear image by a quotient basis; it does not assume that a real semidefinite projection is closed. Indeed the positive semidefinite matrices

$$\begin{pmatrix} 1/n & 1 \\ 1 & n \end{pmatrix}$$

project under $\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right) \mapsto (a, b)$ to $(1/n, 1) \to (0, 1)$, but no matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & c \end{smallmatrix}\right)$ is positive semidefinite: its determinant is -1. This is a counterexample to a weaker replacement argument, not a defect of the source proof.

3 The exact Ram moment stencil

Choose a finite rational basis $w_1, \dots, w_k$ containing all factors of T and write

$$T = \sum_{a,b} C_{ab} w_a \otimes w_b, \quad C = C^T \in \mathrm{Mat}_k(\mathbb{Q}), \quad w_j = (v_j, u_j), \quad U = (u_1 \ \cdots \ u_k).$$

Then $UCU^T = I_d$. For $\Gamma = \mathbb{Z}^k$ and $w(\lambda) = \sum_j \lambda_j w_j$, define $\Lambda_i = \{\lambda \in \Gamma : e_i(w(\lambda)) = 0\}$ and $V_i = \mathbb{Q}\Lambda_i$. Clearing denominators shows that V_i is the rational evaluation kernel. Since $\mathbb{Q}^k/V_i$ injects into $\mathbb{R}$, the tensor represented by C vanishes in $(\mathbb{Q}^k/V_i) \otimes_{\mathbb{Q}} (\mathbb{Q}^k/V_i)$.

Lemma 3.1 (Finite lattice identity; source `stencil:finite`)**.** *For such rational C and subgroups Λ_i, there is a finitely supported $\nu : \mathbb{Z}^k \to \mathbb{Q}$ with*

$$\begin{aligned} &\sum_{\gamma \in q} \nu(\gamma) = 0 \quad (q \in \Gamma/\Lambda_i), \\ &\sum \nu\gamma = 0, \quad \sum \nu\gamma\gamma^T = C. \end{aligned} \tag{3.1}$$

In particular $\sum \nu = 0$.

Proof. In $R = \mathbb{Q}[X_1^{\pm 1}, \dots, X_k^{\pm 1}]$ let $J_i = (X^h - 1 : h \in \Lambda_i)$ and $I = \bigcap_i J_i$. The quotient group-algebra map has kernel J_i: a polynomial of zero coefficient sum in a coset is a sum of differences $c_\gamma(X^\gamma - X^{\gamma_0})$ in this ideal. Complete at $\mathfrak{m} = (X_j - 1)_j$. Formal variables $Y_j = \log X_j$ identify the completion with $\mathbb{Q}[[Y_1, \dots, Y_k]]$. Because $\exp(h \cdot Y) - 1$ equals $h \cdot Y$ times a unit, $J_i \widehat{R} = (h \cdot Y : h \in \Lambda_i)$. The tensor hypothesis places $Q_2 = Y^T C Y/2$ in each of these ideals: the kernel of $\mathrm{Sym}(\mathbb{Q}^k) \to \mathrm{Sym}(\mathbb{Q}^k/V_i)$ is generated by V_i.

The ring R is Noetherian. Completion is flat [7]; tensoring the injection $R/I \to \bigoplus_i R/J_i$ proves $I\widehat{R} = \bigcap_i J_i \widehat{R}$. Write $Q_2 = \sum_{a=1}^N f_a \alpha_a$ with $f_a \in I$ and $\alpha_a \in \widehat{R}$. Replace each α_a by a polynomial b_a in $X_j - 1$ agreeing modulo $\mathfrak{m}^3$. Then $f = \sum f_a b_a \in I$ and $f(e^Y) = Q_2(Y) \pmod{(Y)^3}$. Write $f = \sum \nu(\gamma) X^\gamma$. Membership in I proves the coset cancellations, and the constant, linear, and quadratic terms of $e^{\gamma \cdot Y} = 1 + \gamma \cdot Y + (\gamma \cdot Y)^2/2 + \cdots$ give all the moments. Flatness, not an unsupported truncation of an ideal intersection, is the essential algebraic step. □

3.1 What is genuinely a Ram representation here

Define the signed finite response

$$(\mathcal{R}_{\nu,n} f)(x) = n^2 \sum_\gamma \nu(\gamma) f(x - \gamma/n), \qquad D_C f = \tfrac{1}{2} \sum_{a,b} C_{ab} \partial_a \partial_b f.$$

This is the discrete signed-measure specialization of the actual local moment theorem [18]; it is not a claim about recovering a signal from arbitrary measurements.

Proposition 3.2 (Exact polynomial action and full local remainder)**.** *If $f \in C^3$ on the contributing line segments and $\|D^3 f\| \le M_3$ there, then*

$$|\mathcal{R}_{\nu,n} f(x) - D_C f(x)| \le \frac{M_3}{6n} \sum_\gamma |\nu(\gamma)| \|\gamma\|^3. \tag{3.2}$$

Equality $\mathcal{R}_{\nu,n} f = D_C f$ holds for all polynomials of degree at most two.

Proof. Taylor's formula on $t \mapsto f(x - t\gamma/n)$ has remainder bounded by $M_3 \|\gamma\|^3/(6n^3)$. The mass and first moments cancel, and the second moment is C. Multiply by n^2 and sum absolute remainders. For degree at most two the third derivative is zero. □

The exact evaluation cancellations in (3.1) do not follow from this low-order approximation: they are additional group-algebra identities. A method keeping only the quadratic moment loses precisely the information needed for coordinate-multiset equality.

Proposition 3.3 (Why a positive moment relaxation cannot suffice)**.** *No positive semidefinite real matrix C in the above finite basis can simultaneously have zero ordinary quadratic evaluations at an affinely spanning A and $UCU^T = I_d$. Nor can a nonzero nonnegative ν have zero pushforward on even one quotient Γ/Λ_i.*

Proof. Let $r_i = (e_i(w_j))_j$. If $C \succeq 0$ and $r_i^T C r_i = 0$, then $C r_i = 0$, by a square root of C. The differences $r_i - r_1$ span $\operatorname{im} U^T$ because the differences $a_i - a_1$ span $\mathbb{R}^d$. Hence $CU^T = 0$, contradicting $UCU^T = I_d$. A nonnegative function whose sum is zero on each coset is zero at every lattice point. Thus both tensor and stencil mechanisms intrinsically use signed data. □

Example 3.4 (Indefiniteness already for two points). Take $A = \{0, 1\} \subset \mathbb{R}$, $w_1 = (0, 1)$, $w_2 = (1, -1)$, and

$$C = \begin{pmatrix} 0 & -1/2 \\ -1/2 & 0 \end{pmatrix}, \qquad U = (1, -1).$$

The evaluation rows are $(0,1)$ and $(1,0)$, so their tensor evaluations vanish, while $UCU^T = 1$. The eigenvalues of C are $\pm 1/2$. A corresponding stencil is $-\frac{1}{2}(X_1 - 1)(X_2 - 1)$: its weights at $(0,0), (1,0), (0,1), (1,1)$ are $-1/2, 1/2, 1/2, -1/2$. The mass and first moments vanish, its second moment is C, and summing on every horizontal or vertical fiber gives zero. Those fibers are precisely the two evaluation quotients. Thus the signed mechanism is necessary even in an elementary positive case, not merely in a complicated obstruction.

4 Smoothing, rationalization, and exact two-scale pairs

This section checks source `pairs:bump`, `pairs:rational`, and `pairs:two-scale`. The approximation is confined to producing a strict inequality. Every equality later used against an arbitrary coloring remains exact.

Lemma 4.1 (Signed Gaussian redistribution)**.** *If $C = C^T$ and $UCU^T = I_d$, then for every $\eta > 0$ there is a nonnegative $b \in C_c^\infty(\mathbb{R}^k)$ such that*

$$\int b > 0, \qquad \int (D_C b)_-(x) \|Ux\|^2 \, dx < \eta \int b. \tag{4.1}$$

Proof. Let $H = \operatorname{im} U^T$ and $V = \ker U$. The restriction of C to H is positive definite since $(U^T y)^T C (U^T y) = \|y\|^2$ for $y \neq 0$. Choose $\epsilon > 0$ with $\epsilon \operatorname{tr}(UU^T) < \eta/8$. For sufficiently large L, $S = \epsilon P_H + L P_V$ and $S + C$ are positive definite. Indeed the H-block $A = C_{HH} + \epsilon I$ is positive definite, and the other Schur complement is $C_{VV} + LI - C_{VH} A^{-1} C_{HV} > 0$ for large L. Also $\operatorname{tr}(USU^T) < \eta/8$.

Every $S + tC$, $0 \le t \le 1$, is positive definite. For its centered Gaussian density ϕ_{S+tC}, direct differentiation gives

$$\partial_t \phi_{S+tC} = D_C \phi_{S+tC}.$$

This formula does not assume $C \succeq 0$. The compact covariance interval has uniform positive lower and finite upper eigenvalue bounds; all derivatives have uniform Gaussian decay. Thus $b_* = \int_0^1 \phi_{S+tC} \, dt$ is nonnegative Schwartz, has integral one, and $D_C b_* = \phi_{S+C} - \phi_S$. Consequently

$$\int (D_C b_*)_- \|Ux\|^2 \le \int \phi_S \|Ux\|^2 = \operatorname{tr}(USU^T) < \eta/8.$$

For a smooth cutoff χ_R equal to one on the radius-R ball and zero beyond radius $2R$, put $b_R = \chi_R b_*$. The product rule gives

$$D_C b_R = \chi_R D_C b_* + \sum_{a,b} C_{ab} (\partial_a \chi_R)(\partial_b b_*) + b_* D_C \chi_R.$$

The derivative factors of the cutoff are $O(R^{-1})$ and $O(R^{-2})$; Schwartz decay proves convergence in the weighted norm $\int |\cdot| \|Ux\|^2$. Since the negative-part function is 1-Lipschitz, for large R the negative moment is less than $\eta/4$ and $\int b_R > 1/2$. This proves (4.1). □

Lemma 4.2 (Finite rational weights)**.** *For ν in (3.1) and every $\eta > 0$, there are $n \ge 1$, a finitely supported rational μ, and $\zeta \in \mathbb{Q}_{>0}$ with all quotient pushforwards zero and*

$$\begin{aligned} \sum_\lambda \mu(\lambda) \frac{\lambda \lambda^T}{n^2} &= \zeta C, \\ \sum_\lambda \mu(\lambda)_- \|U\lambda/n\|^2 &< \eta\zeta. \end{aligned} \tag{4.2}$$

Proof. Take b from (4.1). Set $d_n(\lambda) = n^{2-k}b(\lambda/n)$ and $\mu_n = \nu * d_n$. Pushforward commutes with convolution. Substitution $\lambda = \beta + \gamma$ and the first two vanishing moments give, for any finitely supported d,

$$\sum \nu * d = 0, \quad \sum (\nu * d)(\lambda)\lambda = 0, \\ \sum (\nu * d)(\lambda)\lambda\lambda^T = (\sum d)C. \tag{4.3}$$

In particular $\zeta_n = n^{-2}\sum d_n = n^{-k}\sum b(\lambda/n) \to \int b > 0$. Proposition 3.2 gives uniformly

$$n^k \mu_n(\lambda) = D_C b(\lambda/n) + O(n^{-1}).$$

After division by n, all supports lie in one fixed ball with $O(n^k)$ lattice points. The negative-part Lipschitz bound and the bounded weight $\|Ux\|^2$ on this ball give

$$\sum \mu_n(\lambda)_- \|U\lambda/n\|^2 = n^{-k}\sum (D_C b(\lambda/n))_- \|U\lambda/n\|^2 + O(n^{-1}).$$

Riemann sums and (4.1) prove the strict inequality for a large fixed n. Now fix the finite support of d_n. On this finite-dimensional space, $d \mapsto \nu * d$, $d \mapsto n^{-2}\sum d$, and the weighted negative moment are continuous. Approximate d_n by rational d closely enough to retain both strict inequalities and put $\mu = \nu * d$. The exact equations (4.3) and the quotient zeros hold for every d; they have not been approximated. □

Proposition 4.3 (Quantitative Ram rounding budget)**.** *Fix n, a finite support E, and real coefficients d_0 on E. Write $\zeta_0 = n^{-2}\sum d_0 > 0$, $F(d) = \sum_\lambda (\nu * d)(\lambda)_- \|U\lambda/n\|^2$, and suppose $\Delta = \eta\zeta_0 - F(d_0) > 0$. Define*

$$L_E = \max_{\beta \in E} \sum_{\gamma \in \operatorname{supp}\nu} |\nu(\gamma)| \|U(\beta+\gamma)/n\|^2.$$

Every rational d supported on E with

$$\|d - d_0\|_1 < \min\left\{ n^2\zeta_0, \frac{\Delta}{L_E + \eta/n^2} \right\} \tag{4.4}$$

has $\zeta(d) > 0$ and $F(d) < \eta\zeta(d)$ while retaining all exact cancellation and moment identities.

Proof. The negative-part map is 1-Lipschitz, so after interchanging two finite sums,

$$|F(d) - F(d_0)| \le L_E \|d - d_0\|_1, \qquad |\zeta(d) - \zeta_0| \le n^{-2}\|d - d_0\|_1.$$

Therefore $\eta\zeta(d) - F(d) \ge \Delta - (L_E + \eta/n^2)\|d - d_0\|_1 > 0$. The other bound in (4.4) makes $\zeta(d) > 0$. Equation (4.3) proves the exact assertions. If U is known only through numerical approximations, upper bounds for L_E and certified lower bounds for ζ_0, Δ are still required. The proposition does not infer them from floating residuals. □

Theorem 4.4 (Exact two-scale coordinate permutations)**.** *For every $\eta > 0$, a tensor certificate gives finite vectors $f_i, g_i \in \mathbb{R}^\ell$ with*

$$\|f_i - f_j\|^2 = \|a_i - a_j\|^2, \quad \|g_i - g_j\|^2 = q\|a_i - a_j\|^2, \quad q = \frac{\eta}{1+\eta},$$

such that each g_i is a coordinate permutation of f_i.

Proof. Clear denominators in μ by an integer $M > 0$. Form a positive list with $M\mu(\lambda)_+$ copies of $w(\lambda/n)$ and a negative list with $M\mu(\lambda)_-$ copies. Their lengths agree by zero mass. For each i, equality of evaluations is equivalence modulo Λ_i. Zero pushforward says that the two

evaluated lists have exactly the same multiset, not just matching moments. Divide both lists by $\sqrt{M\zeta}$. Their gradient Gram matrices are

$$B_\pm = \zeta^{-1} \sum_\lambda \mu(\lambda)_\pm (U\lambda/n)(U\lambda/n)^T, \quad B_+ - B_- = I_d, \quad \operatorname{tr} B_- < \eta.$$

As $B_- \succeq 0$, $P_0 = \eta I_d - B_- \succeq 0$. Append the same affine rows $(0, r_j)$ to both lists, where $\sum_j r_j r_j^T = P_0$. Their Grams become $(1+\eta)I_d$ and ηI_d. Divide both lists by $\sqrt{1+\eta}$ and evaluate them at each a_i. Common rows and common scaling preserve the coordinate multisets. Affine constant terms cancel in point differences, so the two Gram identities give the claimed distances exactly. □

5 Noncommutative endpoint correction

Let $\mathcal{E} = \mathbb{R}^{(\mathbb{N})}$ and let K be the free group on symbols ξ_z, $z \in \mathcal{E}$. In K^s a path consists of diagonal factors $\Delta g = (g, \dots, g)$ of weight zero and copy factors $X_b = (\xi_{b_1}, \dots, \xi_{b_s})$ of weight $u \geq 0$ when $\|b_i - b_j\|^2 = u\|a_i - a_j\|^2$. Its endpoint is the ordered product; its weight is the sum of the weights of the *chosen expression*. It is not an invariant of the endpoint.

Let M be the monoid generated by diagonals and unit-weight copies. The degree of a represented member is its number of copy factors. Put $N = \ker(K \to \mathbb{Z})$, where every ξ_z maps to 1.

Lemma 5.1 (Pair projections; source `groups:pairs`)**.** *For $i \neq j$, the projection M_{ij} is a group containing $N \times N$.*

Proof. Write $r = \|a_i - a_j\| > 0$. Every ordered pair z, z' at distance r extends to a congruent copy of A in a finite coordinate subspace: extend its direction to an orthonormal frame and translate. For $x = \xi_z$, $y = \xi_{z'}$, both (x, y) and (y, x) are in M_{ij} and

$$(x, y)^{-1} = (x^{-1}, x^{-1})(y, x)(y^{-1}, y^{-1}).$$

Together with the diagonal inverses this makes M_{ij} a group. It contains $(xy^{-1}, 1)$ and $(1, yx^{-1})$ and all their diagonal conjugates. The fixed-distance graph on $\mathcal{E}$ is connected: subdivide any displacement into pieces of length at most $2r$; each pair of endpoints has a common distance-r neighbor, using a perpendicular extra coordinate. The normal closure of all edge differences therefore identifies every free generator and is exactly N. This proves the claim. □

For $g \in K$ let $g^{[i]}$ be supported in coordinate i. Define $L_i = \{g : g^{[i]} = uv^{-1},\ u, v \in M\}$ and let $c_i(g)$ be the least sum of represented degrees in such a right fraction.

Lemma 5.2 (Supported fractions; source `groups:fractions`)**.** *Each L_i is normal in K. Its cost is symmetric, conjugation invariant, and subadditive. A tuple (g_i) with $g_i \in L_i$ has a right fraction of cost at most $\sum_i c_i(g_i)$. Moreover*

$$N^{(s-2)} \subset L_i, \qquad N^{(0)} = N, \quad N^{(r+1)} = [N^{(r)}, N^{(r)}]. \tag{5.1}$$

Proof. Interchanging u, v gives inversion. Left multiplying both by Δk gives conjugation without degree cost. Applying the reverse operations proves equality of costs, not only an inequality. For a right fraction uv^{-1} and $g \in L_i$, choose $h = v_i^{-1} g v_i$ and $h^{[i]} = ab^{-1}$ of minimal cost. Then

$$uv^{-1}g^{[i]} = uh^{[i]}v^{-1} = (ua)(vb)^{-1}. \tag{5.2}$$

The degrees add; this proves subgroup closure and subadditivity and, by iteration, the tuple assertion.

For the induction in (5.1) we need a second, nonquantitative rule. Let $z = cd^{-1}$ be supported on a pair J. For any uv^{-1}, the group property of the projection supplies $b \in M$ such that $t = vb$ is identity on J. Since z and t commute by disjoint support,

$$uv^{-1}z = ubt^{-1}z = ubzt^{-1} = (ubc)(td)^{-1}. \tag{5.3}$$

No cost bound for this auxiliary b is assumed. For $s = 2$ the statement is the preceding lemma. For $s \geq 3$ fix distinct i, j, k. Project away j and use induction to represent any $a \in N^{(s-3)}$ at coordinate i alone in that projection. Lift its numerator and denominator to M, obtaining z_a supported on $\{i, j\}$ with value a at i. Projecting away k similarly gives z_b supported on $\{i, k\}$ with value any $b \in N^{(s-3)}$ at i. The product $z_a z_b z_a^{-1} z_b^{-1}$ is a right fraction by repeated (5.3), and is supported only at i, with value $[a, b]$. Since L_i is a subgroup, it contains $N^{(s-2)}$. Induction applies to all these projections because each contains diagonals and has the same pair-projection hypotheses. □

Let $L = N^{(s-2)}$ and $c(g) = \max_i c_i(g)$ on L. It is finite, conjugation invariant, symmetric, and subadditive.

Lemma 5.3 (Orbit averaging with vanishing relative cost)**.** *Let H be a finite group of coordinate permutations and $Z \subset \mathcal{E}$ a finite H-invariant set. For every $\delta > 0$ there is a tuple $(w_x)_{x \in Z}$ represented by diagonals and factors $(\xi_{hx})_{x \in Z}$, $h \in H$, of positive degree m, such that*

$$w_x w_y^{-1} \in L, \qquad c(w_x w_y^{-1}) \leq \delta m \quad \textit{for } x, y \textit{ in a common orbit.}$$

Proof. Start at $w_x = \xi_x$ of degree one, whose differences lie in N. Suppose the differences on an orbit O lie in $N^{(r)}$. Right-normalize by $w_{x_0}^{-1}$ for $x_0 \in O$. Values on O now belong to $N^{(r)}$. Replace the tuple by

$$w'_x = \prod_{h \in H} w_{hx}$$

in one fixed order. Each orbit point occurs equally often, so modulo the abelian group $N^{(r)}/N^{(r+1)}$ all these products agree. The differences acquire one derived level. On a different orbit, membership in any previously attained normal subgroup is retained: corresponding factors agree modulo that subgroup. Processing finitely many orbits and levels gives differences in L everywhere.

To lower cost, fix O and normalize again so every value there lies in L. Let $B_0 = \max_{x \in O} c(w_x)$, m_0 its represented degree, and $h = |H|$. If $h = 1$ there are no nontrivial orbit differences. Otherwise repeatedly perform

$$w'_x = \left(\prod_{g \in H} w_{gx} \right) w_{x_0}^{-1}, \tag{5.4}$$

using the old value in the final factor. One factor in the product is $a = w_{x_0}$; writing it as PaQ makes $w'_x = P(aQa^{-1})$. Hence the new maximal value cost is at most $(h-1)B_0$, while the represented degree is hm_0. After k steps the ratio of the maximal difference cost to degree is at most

$$\frac{2B_0}{m_0} \left(\frac{h-1}{h} \right)^k \longrightarrow 0.$$

All products of values on this orbit remain in L. On any other orbit, if $a_j b_j^{-1} \in L$, induction on the number of factors using

$$A_j B_j^{-1} = (A_{j-1}(a_j b_j^{-1}) A_{j-1}^{-1})(A_{j-1} B_{j-1}^{-1})$$

gives $c((\prod a_j)(\prod b_j)^{-1}) \leq \sum c(a_j b_j^{-1})$. Thus (5.4) multiplies other difference bounds by at most h while multiplying degree by h. Their ratios do not increase. Treat each orbit in succession to achieve δ on all of them. □

Proposition 5.4 (Endpoint correction; source `groups:endpoints`). *The pairs in Theorem 4.4 give, for every $\delta > 0$, equal-endpoint paths of weights $\lambda > 0$, $\mu \geq 0$ with $\mu/\lambda \leq q + s\delta$.*

Proof. Let H permute the ℓ coordinates of the pairs, and let Z be the union of their finite orbits. Restrict the expression from Lemma 5.3 to f_i and to g_i. The endpoints $D_+ = (w_{f_i})$ and $D_- = (w_{g_i})$ have weights m and qm. Each $r_i = w_{f_i} w_{g_i}^{-1}$ lies in L with cost at most δm. The corresponding component of $D_+^{-1} D_-$ is $w_{f_i}^{-1} r_i^{-1} w_{f_i}$ and has the same bound. Lemma 5.2 gives $D_+^{-1} D_- = uv^{-1}$ of total cost at most $s\delta m$. Append u to D_+ and v to D_-. The endpoints agree because $D_+ u = D_- v$, and their weights are $\lambda = m + \deg u$, $\mu = qm + \deg v$. Therefore $\mu/\lambda \leq q + s\delta$. □

Audit boundary. The proof never assumes that MM^{-1} is a group or that arbitrary fractions can be multiplied. Equations (5.2) and (5.3) provide exactly the supported products required. Degrees are recorded expression lengths even after algebraic word cancellation. No uniform bound in η, δ, s, or the number of colors has been established by this finite-cost argument.

6 Finite synchronization and arbitrary-color forcing

Lemma 6.1 (A finite list of exact weights). *Suppose two paths have one endpoint and weights $\lambda > 0$ and $\mu = \rho\lambda$ with $0 \leq \rho < 1$. Given a finite list $\beta_j \in [\rho L_0, L_0]$, $L_0 > 0$, one can construct a single endpoint admitting paths of every weight β_j.*

Proof. Set $\alpha_j = (\beta_j/L_0 - \rho)/(1 - \rho) \in [0, 1]$. Partition $[0, 1]$ at all these numbers, discarding zero-length pieces. For each interval of length l, apply to both original paths the endomorphism $\xi_z \mapsto \xi_{\sqrt{L_0 l/\lambda}\, z}$. Their endpoints still agree and their weights are $L_0 l$ and $\rho L_0 l$. Concatenate intervals in a fixed order. For target j choose the larger weight before α_j and the smaller after it. The endpoint is independent of these choices and the weight is $L_0[\alpha_j + \rho(1 - \alpha_j)] = \beta_j$. This requires neither commutativity nor a rationality assumption on the weights. □

The source's `paths:synchronization` is the specialization $L_0 = 1$, $\beta_j = 1/j$, $1 \leq j \leq n$, under $\rho < 1/n$. Our formulation records the finite-list extension explicitly.

6.1 A common-factor realization theorem

Let $\mathcal{S}$ be the monoid of finite strings on alphabet $\mathcal{E}$, with identity the empty string. For $B \subset \mathcal{S}$, ultrafilters multiply by

$$B \in uv \quad \Longleftrightarrow \quad \{x : x^{-1}B \in v\} \in u, \qquad x^{-1}B = \{y : xy \in B\}. \tag{6.1}$$

This is a preimage notation, not inversion of a string.

Lemma 6.2 (The corner group). *The compact space $\beta\mathcal{S}$ is a right-topological semigroup with identity δ_ϵ and has an idempotent p for which $p(\beta\mathcal{S})p$ is a group with identity p.*

Proof. Ultrafilter axioms define a closed subset of $\{0, 1\}^{\mathcal{P}(\mathcal{S})}$, so compactness follows from the product theorem. The membership test in (6.1) preserves finite intersections and complements, hence defines an ultrafilter. Associativity is the identical nested membership condition $\{x : \{y : \{z : xyz \in B\} \in w\} \in v\} \in u$. For fixed v, the inverse image of the clopen $\overline{B}$ under $u \mapsto uv$ is clopen; right translations are continuous.

By intersection of chains choose a minimal nonempty compact left ideal L_0. For every $v \in L_0$, the compact left ideal $(\beta\mathcal{S})v$ equals L_0. Choose a minimal nonempty compact subsemigroup

$H_0 \subset L_0$. For $x \in H_0$, H_0x is again such a subsemigroup: $(ax)(bx) = a(xb)x \in H_0x$. Hence $H_0x = H_0$. The nonempty compact subsemigroup $\{y \in H_0 : yx = x\}$ then equals H_0, proving $x^2 = x$. Choose such an idempotent p. The corner $G = p(\beta\mathcal{S})p$ is a monoid. Every $y \in G$ lies in L_0, so $vy = p$ for some v. Then $(pvp)y = p$, giving a left inverse in G. In any monoid in which every element has a left inverse, $ay = p$, $ba = p$ imply $b = b(ay) = (ba)y = y$, whence $ya = p$ as well. Thus G is a group. This supplies the standard semigroup step directly; compare Ellis and Hindman–Strauss [8, 9]. □

Lemma 6.3 (Simultaneous realization; source `paths:realization`)**.** *Suppose $B \in q_{i,1} \cdots q_{i,M}$ for $1 \le i \le s$, where each column either consists of the same ultrafilter in all roles or consists of principal ultrafilters at prescribed strings $z_{i,j}$. There are strings $x_{i,j}$ whose row products belong to B, with common strings in common columns and $x_{i,j} = z_{i,j}$ in the prescribed columns.*

Proof. At column j keep prefixes P_i with $P_i^{-1}B \in q_{i,j} \cdots q_{i,M}$. Put $v_i = q_{i,j+1} \cdots q_{i,M}$, using δ_ϵ for an empty product. Then $E_i = \{x : x^{-1}(P_i^{-1}B) \in v_i\} \in q_{i,j}$. For a common column choose one string in the finite intersection $\bigcap_i E_i$, which belongs to the common ultrafilter. For a principal column the prescribed string is in E_i. Update each prefix and continue. After the last column the invariant says $P_i \in B$. □

Theorem 6.4 (Path forcing; source `paths:criterion`)**.** *If for every $n \ge 1$ there are equal-endpoint paths with $\lambda > 0$ and $0 \le \mu/\lambda < 1/n$, then A is Ramsey.*

Proof. Fix r and any coloring $c : \mathcal{E} \to [r]$. Use the classical finite Hales–Jewett theorem [5, 6] to choose n such that every r-coloring of $[s]^n$ has a combinatorial line with a nonempty variable-position set. Apply Lemma 6.1 to obtain an endpoint $w \in K^s$ with representations of all weights $1, 1/2, \ldots, 1/n$ *before coloring the words.*

Choose disjoint infinite coordinate blocks and linear isometries $J_j : \mathcal{E} \to \mathcal{E}$ into them. Map a string $(z_1, \ldots, z_L)$ to $\Psi(z) = \sum_{j=1}^{L} J_j z_j$. For equal string lengths,

$$\|\Psi(z) - \Psi(z')\|^2 = \sum_j \|z_j - z'_j\|^2.$$

Let $B_1, \ldots, B_r$ be the string color classes induced by $c\Psi$. Freeness gives a group homomorphism $\psi : K \to p(\beta\mathcal{S})p$ taking ξ_z to $p\delta_{(z)}p$. Color a word $i_1 \cdots i_n$ by the unique k such that $B_k \in \psi(w_{i_1}) \cdots \psi(w_{i_n})$.

Let the Hales–Jewett line have t variable positions. At each variable position expand the same chosen path representation of w with weight $1/t$. Diagonals give common ultrafilters; a copy gives $p\delta_{(b_i)}p$, whose outer factors are common and whose middle factor is a prescribed one-letter string. Fixed positions are common. An empty group product is retained as the common factor p, since the corner-group identity is p, not δ_ϵ. Simultaneous realization gives strings in the same B_k. Their common factors have identical lengths and entries, and the other factors all have length one. They therefore align, and their squared distances are

$$t\,(1/t)\|a_i - a_j\|^2 = \|a_i - a_j\|^2.$$

Thus every coloring of $\mathcal{E}$ has the required copy.

Finally, if for this r every finite $\mathbb{R}^D$ admitted an avoiding coloring, consider the compact space $[r]^{\mathcal{E}}$. For each congruent copy impose the clopen condition that its colors are not all equal. Every finite set of conditions lies in one finite coordinate subspace, so it is satisfiable. Compactness gives an avoiding coloring of $\mathcal{E}$, a contradiction. This proves existence of a finite $D(A, r)$. □

Theorem 6.5 (Reconstructed classification, with declared inputs)**.** *Assuming the classical finite Hales–Jewett theorem, Noetherian completion flatness, and ZFC,* (2.1) *is equivalent to* (1.1) *for every finite nonempty configuration in its affine span, treating a singleton separately.*

Proof. Necessity is Theorem 2.3 and exact descent. For sufficiency and any n, choose $\eta > 0$ with $q = \eta/(1+\eta) < 1/(2n)$. The finite-stencil and two-scale results give the pair of copies. Proposition 5.4 with $\delta = 1/(2sn)$ gives equal endpoints of weight ratio less than $1/n$. Theorem 6.4 applies. The order of choices is

$$\begin{aligned} r \;&\longrightarrow\; n_{\mathrm{HJ}} \;\longrightarrow\; (q, \delta, w, \{1/t\}_{t=1}^{n_{\mathrm{HJ}}}) \\ &\longrightarrow\; \text{word coloring} \;\longrightarrow\; t \;\longrightarrow\; \text{exact copy.} \end{aligned}$$

There is no selection of a new endpoint after seeing t, and no limit of monochromatic approximate copies. □

7 Geometric consequences, positive and negative

7.1 Finite transitive hosts

Proposition 7.1 (Source `cons:transitive`)**.** *Every nonempty subtransitive configuration satisfies the tensor criterion. Its Ramsey conclusion follows from Theorem 6.5.*

Proof. Embed A as (y_i) in a finite transitive set $Y \subset \mathbb{R}^n$, and let H be its full finite isometry permutation group. For each h, the map $a_i \mapsto h(y_i)$ extends to an affine isometry; let w_{hj} be its coordinate rows. Each such row family has gradient Gram I_d. Let $c_{hj} = ((h(y_1))_j, 0)$ be constant rows and set

$$T = \frac{1}{|H|} \sum_{h,j} (w_{hj} \otimes w_{hj} - c_{hj} \otimes c_{hj}).$$

The gradients of the constants vanish, so $G(T) = I_d$. For each i, the multiset $(h(y_i))_{h \in H}$ contains every point of Y equally often. Hence each rational tensor evaluation is zero. The proof constructs a certificate without assuming the transitive-host Ramsey conclusion in advance. □

7.2 Quadratic interpolation and at most five circle points

Proposition 7.2 (Adjugate lift; source `cons:quadratic`)**.** *Suppose A is spherical and the rows $(p_{i\alpha} p_{i\beta})_{\alpha,\beta}$ are independent over F. Then* (2.1) *holds.*

Proof. The ordinary sphere equations and spatial-block equations are linear over F. Their real solution therefore descends by row reduction to $h \in F^{(d+1)^2}$ with spatial block I_d. Let D be the tensor evaluation matrix with entries $p_{i\alpha} \otimes p_{i\beta}$, and lift h to b entrywise by $x \mapsto x \otimes 1$. Select an s-column square submatrix M with $\delta = m(\det M) \neq 0$, possible by row independence. If ι inserts into these columns, then $D\iota = M$ and

$$z = (\det M) b - \iota \operatorname{adj}(M) D b \in \ker D, \qquad m(z) = \delta h.$$

Multiply z by $\delta^{-1} \otimes 1$. Its evaluation is zero and its multiplied spatial block is I_d. No inverse of $\det M$ in B has been used; B need not even be a domain. □

Corollary 7.3 (Source `cons:quadrilateral`)**.** *Every nonempty set of at most five distinct points on a circle satisfies the tensor criterion and, using sufficiency, is Ramsey.*

Proof. Enlarge it to five distinct points on the same circle. For a chosen point pair the other four into two pairs and multiply equations of the two secant lines. Each factor has coefficients in the coordinate field. No three distinct circle points are collinear; the product is nonzero at the chosen point and zero at the other four. Rescaling gives the five quadratic Lagrange evaluation functions. Thus evaluation has rank five, and Proposition 7.2 applies. A subset inherits Ramsey-ness by restricting a monochromatic copy. □

On any circle, restriction of quadratics has dimension five: after a real similarity its span is $1, \cos\theta, \sin\theta, \cos 2\theta, \sin 2\theta$. Equivalently the circle equation is the unique quadratic relation for at least five distinct circle points, since the preceding interpolation gives rank five. Thus for $s \geq 5$ the evaluation rank is exactly five. For six or more points the row-independence test is not available. Its failure is not a tensor obstruction: the algebraic examples in Section 8 can have arbitrarily many points.

For the kite

$$K_a = \{(-1,0), (1,0), (a, \sqrt{1-a^2}), (a, -\sqrt{1-a^2})\}, \quad -1 < a < 1,$$

the positive conclusion just obtained combines, when a is transcendental, with the external non-subtransitivity theorem of Leader–Russell–Walters [4]. Their Corollary 2 is the precise input; it is not proved by the cut-cone barrier. Their parameters are $\alpha = -1$, $\beta = 1 + a$, so the hypothesis of their Theorem 1 applies. The source consequently refutes the necessity direction of their proposed subtransitive characterization [3], subject to the full sufficiency chain. We do not turn this external theorem or its priority into a new Ram result.

7.3 Nine algebraically independent circle parameters

Theorem 7.4 (Source `cons:nine`; necessity alone suffices)**.** *Let $t_1, \ldots, t_9$ be algebraically independent over $\mathbb{Q}$. Then*

$$a_i = \left(\frac{1 - t_i^2}{1 + t_i^2}, \frac{2t_i}{1 + t_i^2} \right)$$

are distinct points on a circle and are not Ramsey.

Proof. The circle equation is immediate. The inverse parameter is $t = y/(1 + x)$, so $F = \mathbb{Q}(t_1, \ldots, t_9)$. The points are distinct and any three span the circle's plane. Independence identifies B with the localization of $\mathbb{Q}[T_1, \ldots, T_9, U_1, \ldots, U_9]$ by all nonzero polynomials in the T variables alone and in the U variables alone. In particular it is a domain embedded in the joint rational-function field. For $p(t) = (1, (1 - t^2)/(1 + t^2), 2t/(1 + t^2))^T$, the nine tensor equations have rows $p(T_i)^T \otimes p(U_i)^T$ in a 9×9 matrix D. Its determinant is not the zero rational function: specialize its nine argument pairs to the grid $\{0, 1, -1\}^2$, in that order. Its displayed denominators remain nonzero and the matrix becomes $C \otimes C$, where

$$C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix}, \quad \det C = 2, \quad \det(C \otimes C) = 2^6 = 64.$$

This is a specialization of one rational expression, not a homomorphism from the entire localized ring (which inverts polynomials vanishing at the grid). Over the fraction field D is invertible, so $P = 0$ is the only tensor-evaluation solution even over B. It cannot have multiplied spatial block I_2. Exact descent and Theorem 2.3 give the claimed non-Ramsey conclusion. □

The determinant computation does not say that *all* nine-point circle sets fail. Rational or algebraic parameters have relations and are governed by a different tensor ring. The grid used in the proof is not itself a nine-point non-Ramsey configuration.

7.4 Twelve points and a one-parameter derivation obstruction

Let $t = \sum_{m\geq 1} 10^{-m!}$, $u = (1-t^2)/(1+t^2)$, $v = 2t/(1+t^2)$, $Q_0 = \{(\pm 1, 0), (0, \pm 1)\}$, and

$$R_+ = \begin{pmatrix} u & -v \\ v & u \end{pmatrix}, \qquad R_- = \begin{pmatrix} u & v \\ -v & u \end{pmatrix}.$$

Proposition 7.5 (Source `cons:twelve`). *$R_+Q_0 \cup R_-Q_0 \cup Q_0$ has twelve distinct circle points and fails the tensor criterion, hence is not Ramsey.*

Proof. Partial sums $t_m = p_m/q_m$, $q_m = 10^{m!}$, satisfy $0 < t - t_m < 2q_m^{-(m+1)}$ for $m \geq 2$. If t were algebraic with irreducible $P \in \mathbb{Z}[T]$ of degree d, then for all sufficiently large m, $P(t_m) \neq 0$, $|P(t_m)| \geq q_m^{-d}$, while the mean value bound gives $|P(t_m)| < 2Cq_m^{-(m+1)}$, a contradiction. The possible single rational root equality does not affect the sufficiently large m argument. Also $0.11 < t < 0.12$; the angles $0, \pm 2\arctan t$ differ by nonzero amounts less than $\pi/2$. Thus the three squares are disjoint.

The coordinate field is $F = \mathbb{Q}(t)$, since $t = v/(1+u)$. The derivation $\partial = (1+t^2)d/dt\,/2$ obeys $\partial u = -v$, $\partial v = u$. Give weights $\beta = 1$ to the two rotated squares and $\beta = -2$ to the fixed square, and set $p_i = (1, x_i)^T$. Because $\sum_{Q_0} q = 0$, $\sum_{Q_0} qq^T = 2I_2$, and $\partial R_+ = R_+J$, $\partial R_- = -R_-J$, where $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, we have

$$\begin{aligned} \sum_i \beta_i p_i p_i^T &= 0, \\ \sum_i \beta_i (\partial p_i) p_i^T &= 0, \\ \sum_i \beta_i (\partial p_i)(\partial p_i)^T &= \operatorname{diag}(0, 4, 4). \end{aligned} \tag{7.1}$$

The respective spatial blocks are $2(1+1-2)I_2$, $2J-2J$, and $2I_2+2I_2$; constant and first-moment blocks also vanish. Define the rational-linear map $\Phi : F \otimes_{\mathbb{Q}} F \to F$ by $\Phi(a \otimes b) = (\partial a)(\partial b)$. Expanding one entry $P_{\alpha\beta} = a \otimes b$, the product rule in $\partial(p_{i\alpha}a)\partial(p_{i\beta}b)$ gives four terms. The terms with two coefficient derivatives vanish by the first identity in (7.1), and the two mixed terms by the second and its transpose. The remaining term gives

$$\sum_i \beta_i \Phi(E_i(P)) = 4 \sum_{\alpha=1}^{2} m(P_{\alpha\alpha}).$$

A certificate would make the two sides 0 and 8. This contradiction uses only a derivation on F, not any regularity assumption on colorings. □

The checker may specialize the displayed rational-function moment identities at rational t to test their algebra. Such a specialization does *not* give a derivation of $\mathbb{Q}$ with the displayed derivative values and does not prove non-Ramseyness at rational t.

7.5 A derivation bridge and seven-point refinement

The coefficient choice in the following seven-point theorem is from Pálvölgyi [15, Appendix, Theorem A.3]. That appendix is explicitly labeled unchecked by its author. We verify this particular claim below, using a direct tensor obstruction in place of its real-coefficient Rado input. This is not a validation of the entire appendix or a novelty claim.

Proposition 7.6 (Derivation-to-tensor obstruction)**.** *Let $F \subset \mathbb{R}$ contain the coordinates of A, let $D : F \to F$ be a derivation, and let $\lambda_i \in F$. With $p_i = (1, a_i)^T$, suppose*

$$S = \sum_i \lambda_i p_i p_i^T = 0, \qquad M = \sum_i \lambda_i (Dp_i) p_i^T = M^T.$$

Put $N = \sum_i \lambda_i (Dp_i)(Dp_i)^T$. If $\sum_{\alpha=1}^{d} N_{\alpha\alpha} \neq 0$, there is no tensor certificate over $F \otimes_{\mathbb{Q}} F$, and A is not Ramsey. The weights need not be annihilated by D.

Proof. A derivation annihilates $\mathbb{Q}$, so $\Phi(a \otimes b) = (Da)(Db)$ defines a rational-linear map $F \otimes_{\mathbb{Q}} F \to F$. For a single entry $P_{\alpha\beta} = a \otimes b$, expand

$$\begin{aligned} D(p_{i\alpha} a) D(p_{i\beta} b) = {} & p_{i\alpha} p_{i\beta} (Da)(Db) + (Dp_{i\alpha}) p_{i\beta} a (Db) \\ & + p_{i\alpha} (Dp_{i\beta}) (Da) b + (Dp_{i\alpha})(Dp_{i\beta}) ab. \end{aligned}$$

Multiply by λ_i and sum after applying Φ; no derivative is applied to a weight. The first term vanishes by $S = 0$. Symmetry of M combines the middle two terms into $M_{\alpha\beta} D(ab)$. Extending by additivity to all entries gives

$$\sum_i \lambda_i \Phi(E_i(P)) = \sum_{\alpha,\beta=0}^{d} \big(N_{\alpha\beta} m(P_{\alpha\beta}) + M_{\alpha\beta} D(m(P_{\alpha\beta})) \big). \tag{7.2}$$

Since $Dp_{i0} = 0$, the constant row of M is zero; symmetry also annihilates its constant column. Both constant blocks of N are zero. For a certificate, the spatial entries of $m(P)$ are the rational constants $\delta_{\alpha\beta}$. Thus the right side of (7.2) is $\sum_{\alpha=1}^{d} N_{\alpha\alpha} \neq 0$, whereas the left side is zero. If F is larger than the coordinate field, a certificate over that smaller field would map into $F \otimes_{\mathbb{Q}} F$, so is also excluded. Necessity proves the non-Ramsey conclusion. □

Theorem 7.7 (Seven-point consequence of Appendix A.3)**.** *Fix three distinct real numbers t_1, t_2, t_3. If t_4, t_5, t_6, t_7 are algebraically independent over $K = \mathbb{Q}(t_1, t_2, t_3)$, then the seven points*

$$a_i = (x_i, y_i) = \left(\frac{1 - t_i^2}{1 + t_i^2}, \frac{2t_i}{1 + t_i^2} \right)$$

fail the tensor criterion and are not Ramsey.

Proof. All parameters are distinct, the points lie on the unit circle, and $t_i = y_i/(1 + x_i)$ identifies their coordinate field with $F = K(t_4, t_5, t_6, t_7)$. Put

$$H(T) = \prod_{i=1}^{7} (T - t_i), \qquad g(T) = \prod_{i=1}^{3} (T - t_i).$$

All $H'(t_i)$ are nonzero. Lagrange interpolation for a polynomial f of degree at most six, followed by comparison of its T^6 coefficient, gives

$$\sum_{i=1}^{7} \frac{f(t_i)}{H'(t_i)} = [T^6] f(T). \tag{7.3}$$

Indeed the interpolation basis is $H(T)/((T - t_i)H'(t_i))$, whose leading coefficient is $1/H'(t_i)$.

Algebraic independence permits the K-derivation $D = \sum_{i=4}^{7} g(t_i) \partial/\partial t_i$ on F, with the quotient rule on rational functions. It satisfies $Dt_i = g(t_i)$ for every i, including the first three where both sides are zero. Define

$$\lambda_i = \frac{(1 + t_i^2)^2}{H'(t_i)}, \qquad b_i = \frac{2g(t_i)}{1 + t_i^2}, \qquad J(x, y) = (-y, x).$$

Direct differentiation gives $Da_i = b_i J a_i$. Every entry of $\lambda_i p_i p_i^T$ is $h(t_i)/H'(t_i)$ for a polynomial h of degree at most four. Equation (7.3) therefore gives $S = 0$. It also gives

$$\sum_i \lambda_i b_i(1, x_i, y_i) = 2\sum_i \frac{g(t_i)}{H'(t_i)}(1 + t_i^2, 1 - t_i^2, 2t_i) = 0,$$

since each numerator has degree at most five. Thus $\sum_i \lambda_i Da_i = 0$, making the constant column of M zero, and

$$M_{21} - M_{12} = \sum_i \lambda_i b_i(x_i^2 + y_i^2) = \sum_i \lambda_i b_i = 0.$$

The constant row is automatically zero, so M is symmetric. Finally, since g^2 is monic of degree six,

$$\sum_{\alpha=1}^{2} N_{\alpha\alpha} = \sum_i \lambda_i \|Da_i\|^2 = \sum_i \lambda_i b_i^2 = 4\sum_i \frac{g(t_i)^2}{H'(t_i)} = 4.$$

Proposition 7.6 applies. Only a derivation on the coordinate field has been used; extension to all real numbers and the real-coefficient inhomogeneous coloring theorem are unnecessary. □

This argument strengthens the generic nine-point obstruction while leaving its independently checked determinant proof intact. It proves neither that all seven-point circle sets fail nor any six-point classification. Rational samples may verify the displayed interpolation identities, but do not realize the prescribed nonzero derivation on $\mathbb{Q}$; algebraic independence remains a genuine theorem hypothesis.

8 Algebraic-coordinate extensions and exact decision data

Theorem 8.1 (Separability-idempotent lift)**.** *Let $F \subset \mathbb{R}$ be a finite extension of $\mathbb{Q}$ containing all coordinates of A. Then A is spherical if and only if it has a certificate over $F \otimes_{\mathbb{Q}} F$. Thus, conditional on the reconstructed sufficiency theorem, every finite spherical configuration with real algebraic coordinates is Ramsey.*

Proof. Choose a primitive element α of the characteristic-zero finite extension. For completeness, simplicity follows by taking successive linear combinations of finitely many generators: among the finitely many distinct embeddings into an algebraic closure, only finitely many rational coefficients can make two such linear combinations have the same image. Avoid these coefficients; the resulting element has $[F : \mathbb{Q}]$ distinct images and generates F. Let $f \in \mathbb{Q}[Y]$ be the minimal polynomial and identify

$$F \otimes_{\mathbb{Q}} F \simeq F[Y]/(f(Y)), \quad x \otimes 1 \leftrightarrow x, \quad 1 \otimes \alpha \leftrightarrow Y.$$

Factor $f(Y) = (Y - \alpha)g(Y)$ in $F[Y]$. Separability gives $g(\alpha) = f'(\alpha) \neq 0$. Set

$$e = \frac{g(Y)}{g(\alpha)} \pmod{f}. \tag{8.1}$$

Then $m(e) = 1$, $(Y - \alpha)e = 0$, and $e^2 = e$. Indeed modulo $Y - \alpha$, $e = 1$, while modulo g, $e = 0$; the factors are coprime. For every $x \in F$, expressed as a polynomial in α,

$$(x \otimes 1)e = (1 \otimes x)e. \tag{8.2}$$

If A is spherical, its sphere equation and identity spatial block form a consistent linear system over F, so choose an F-valued solution H. Define $P_{\alpha\beta} = e(H_{\alpha\beta} \otimes 1)$. Using (8.2),

$$E_i(P) = e((p_i^T H p_i) \otimes 1) = 0, \qquad m(P_{\alpha\beta}) = H_{\alpha\beta}.$$

This is the required certificate. The converse is the sphere equation already proved. Using a larger finite field causes no problem: exact descent through the affine tensor formulation returns a certificate in the actual coordinate field. □

For $F = \mathbb{Q}(\sqrt{2})$, writing $a = \sqrt{2}$, the idempotent is $e = 1/2 + (a \otimes a)/4$. Direct reduction by $a^2 = 2$ verifies $e^2 = e$, $m(e) = 1$, and $(a \otimes 1 - 1 \otimes a)e = 0$. For $F = \mathbb{Q}$, simply take $e = 1$; rational sphere equations already are tensor equations. These are elementary separability consequences, not a claimed new separability theorem or a novelty assertion about algebraic Euclidean Ramsey sets.

Proposition 8.2 (Finite number-field certificate alternative)**.** *Given an exact number-field presentation, the tensor equations are a finite rational linear system. Either a rational solution or a rational left-null inconsistency witness can be verified by exact arithmetic.*

Proof. Choose a rational basis $b_1, \ldots, b_h$ with $b_1 = 1$ and exact multiplication constants $b_a b_b = \sum_c \mu_{ab}^c b_c$. Expand every entry of P in the basis $b_a \otimes b_b$. Multiplication by each coordinate on either side of B is a rational matrix from this multiplication table. Expanding $E_i(P) = 0$ and the multiplied spatial block therefore gives $Az = b$ over $\mathbb{Q}$, with $(d+1)^2h^2$ unknowns, at most $sh^2 + d^2h$ scalar rows. Gaussian elimination either gives a solution or a row vector y with $y^T A = 0$, $y^T b = 1$, after normalization of a nonzero inconsistent row. Both outputs are checked by multiplication. This alternative assumes the field presentation and coordinate expressions are already correct; it does not recognize them from rounded coordinates. □

For a general finitely generated coordinate field, $F_0 = \mathbb{Q}[(a_i)_\alpha]$ is a finitely generated domain and B is a localization of the Noetherian F-algebra $F \otimes_{\mathbb{Q}} F_0$. Its evaluation kernel is finitely generated as a B-module. The multiplied spatial image is the F-span of the multiplied blocks of any generating set, since each $c \in F$ lifts to $c \otimes 1$. This is a finite algebraic description *relative to exact relations*, not a numerical decision algorithm for arbitrary real inputs. The finite number-field system above is a concrete special case where all required field arithmetic is supplied.

Corollary 8.3 (Dense opposite outcomes for at least seven circle points)**.** *On the open parameter space of ordered distinct stereographic circle parameters $(t_1, \ldots, t_s)$, $s \geq 7$, both Ramsey and non-Ramsey configurations are dense, using sufficiency for the positive statement. Consequently no positive uniform distance tolerance on coordinates or squared distances decides Ramsey-ness throughout this class.*

Proof. Every parameter neighborhood contains distinct rational parameters, whose points have rational coordinates and satisfy the tensor condition by Theorem 8.1. It also contains algebraically independent parameters: choose them successively in their open intervals outside the algebraic closure of the field generated by the preceding choices. That field and its algebraic closure are countable, whereas each interval is uncountable. The first seven points fail Ramsey-ness by Theorem 7.7, and a set containing a non-Ramsey subset is non-Ramsey. The parameterization and all pairwise squared distances are continuous, proving the last assertion. This is not a statement about every configuration of a given cardinality, nor about six points. □

9 The private Ram cut criterion and its exact boundary

The following results retain the applicable mathematics of the actual private report [17], rather than inferring it from its title. Write $q_{ij} = \|a_i - a_j\|^2$ and let $\delta_S(i,j)$ equal 1 when exactly one of i, j belongs to S, and 0 otherwise. Complementary cuts coincide. Set $\mathrm{CUT}_m = \mathrm{cone}\{\delta_S\}$.

Proposition 9.1 (Box equivalence and planar rigidity)**.** *A finite configuration embeds as vertices of a rectangular box exactly when $q \in \mathrm{CUT}_m$. Four distinct coplanar box vertices form a rectangle. The same restriction holds for a vertex subset of a product of regular simplices.*

Proof. A box coordinate of side a_ℓ supplies the cut contribution $a_\ell^2 \delta_{S_\ell}$. Conversely, from $q = \sum w_S \delta_S$, $w_S \geq 0$, construct $y_i = (\sqrt{w_S}\, 1_{i \in S})_S$. Their squared distances are exactly q. Each cut satisfies the triangle inequalities for squared distances; the cosine law therefore makes every constituent angle nonobtuse. Three distinct collinear points are impossible. Nor can one of four coplanar points be inside the other three: the three angles around it sum to 360 degrees, so one is at least 120 degrees. The four points form a convex quadrilateral with all angles at most 90 degrees. Their sum is 360 degrees, forcing a rectangle. A regular simplex has the one-hot realization $\{ae_j\}_j$; concatenating such blocks embeds every product of regular simplices in a box. □

9.1 All four-point cut weights, with their nullspace

In edge order $(12, 13, 14, 23, 24, 34)$ use singleton cuts with weights $a_1, \ldots, a_4$ and balanced cuts $12|34, 13|24, 14|23$ with weights b_1, b_2, b_3. Let

$$S_1 = q_{12} + q_{34}, \quad S_2 = q_{13} + q_{24}, \quad S_3 = q_{14} + q_{23},$$
$$S = S_1 + S_2 + S_3, \qquad D_i = \sum_{j \neq i} q_{ij}.$$

Theorem 9.2 (Exact four-point coordinates; private `thm:weights229`)**.** *The complete affine solution of the cut equations is*

$$a_i = (2D_i - S + t)/4, \quad b_j = (S - 2S_j - t)/4. \tag{9.1}$$

It is nonnegative exactly for $L \leq t \leq U$, where $L = \max_i(S - 2D_i)$ and $U = S - 2\max_j S_j$. For nonnegative edge data, the triangle inequalities imply $0 \leq L \leq U$. Thus $\mathrm{CUT}_4 = \mathrm{MET}_4$, and either endpoint of this interval gives at most six nonzero cut coordinates. The cut dictionary has nullspace $\mathbb{Q}(1, 1, 1, 1, -1, -1, -1)$.

Proof. Writing $t = \sum_i a_i$, the edge equations give $D_i = 2a_i + t + 2\sum_j b_j$ and $S = 3t + 4\sum_j b_j$. Solving gives the formula for a_i; opposite-edge sums give the formula for b_j. Conversely substitution, for example $a_1 + a_2 + b_2 + b_3 = q_{12}$, verifies every edge after relabeling. Hence these are all solutions, and their common one-dimensional direction is the displayed nullspace vector. The average of $S - 2D_i$ is zero, so $L \geq 0$. For every i, j, $D_i \geq S_j$ follows from a triangle inequality after cancelling the common incident edge. Thus $L \leq U$. Every cut satisfies triangle inequalities, proving the reverse inclusion. At L a singleton weight vanishes and at U a balanced weight vanishes. □

9.2 Exact path repair and the robust dual margin

For nonnegative complete edge data let

$$\eta(q) = \max_{i<j} \max_{P: i \rightsquigarrow j \text{ simple}} \frac{(q_{ij} - \sum_{e \in P} q_e)_+}{|P| + 1}. \tag{9.2}$$

Theorem 9.3 (Private `thm:repair229`)**.** $\mathrm{dist}_\infty(q, \mathrm{MET}_m) = \eta(q)$*. A minimizer is the shortest-path pseudometric with edge lengths $q_e + \eta(q)$. For $m = 4$, but not in general for $m \geq 5$, this is also distance to CUT_m.*

Proof. If $d \in \mathrm{MET}_m$ and $\|d-q\|_\infty \le h$, then for every simple path

$$q_{ij} - h \le d_{ij} \le \sum_{e\in P} d_e \le \sum_{e\in P} q_e + |P|h.$$

Thus $h \ge \eta(q)$. Conversely put $h = \eta(q)$ and take the shortest paths for $q_e + h \ge 0$. The direct edge bounds the distance above by $q_{ij} + h$. Every simple path has length at least $q_{ij} - h$ by (9.2); cycles can be removed without increasing length. This gives the lower bound. Shortest paths form a nonnegative pseudometric. For $m = 4$ use $\mathrm{CUT}_4 = \mathrm{MET}_4$. □

For four points there are 24 nontrivial simple-path tests: two paths of length two and two of length three for each of six endpoint pairs. For positions $0, 1, 2, 3$ on a line, $q = (1, 4, 9, 1, 4, 1)$ gives $\eta = 3/2$, strictly larger than the triangle-only bound $4/3$. The repair $(5/2, 5, 15/2, 5/2, 5, 5/2)$ has singleton weights $a_1 = a_4 = 5/2$, $a_2 = a_3 = 0$, and balanced weight $b_1 = 5/2$.

If A_{cut} is the cut dictionary and $A_{\mathrm{cut}}^T y \ge 0$, $y^T q < 0$, then

$$\mathrm{dist}_\infty(q, \mathrm{CUT}_m) \ge \frac{-y^T q}{\|y\|_1}. \tag{9.3}$$

Indeed $y^T d \ge 0$ for every cut vector d, and $-y^T q \le y^T(d-q) \le \|y\|_1 \|d-q\|_\infty$. A violated squared triangle inequality of size γ gives $\gamma/3$. More generally every nontrivial normalized path functional in (9.2) has dual norm one, so $|\eta(q) - \eta(\widetilde{q})| \le \|q - \widetilde{q}\|_\infty$. With certified measurement error e, the four-point cut distance lies in $[\max(0, \eta(\widetilde{q}) - e), \eta(\widetilde{q}) + e]$. These are geometric error bars, not a tolerance for exact congruence under an arbitrary coloring.

9.3 Two explicit separations from the box route

Example 9.4 (A rational cyclic kite). Take $(-1, 0), (1, 0), (3/5, 4/5), (3/5, -4/5)$. In the above edge order,

$$q = (4, 16/5, 16/5, 4/5, 4/5, 64/25).$$

Its tensor certificate is $P = \mathrm{diag}(-1, 1, 1)$ over $\mathbb{Q} \otimes \mathbb{Q}$. Yet $q_{34} - q_{23} - q_{24} = 24/25$, so it cannot be box-embedded. The 24 path tests give $\eta = 8/25$, and the exact cut repair is

$$d = (108, 88, 88, 28, 28, 56)/25.$$

It obeys the triangle inequalities and has sup error $8/25$; (9.1) supplies a cut representation. The triangle witness proves optimality. This configuration is tensor-feasible and, under sufficiency, Ramsey, despite a strictly positive cut distance. It is not asserted to be non-subtransitive.

Proposition 9.5 (Five circle points exceed the box-vertex route)**.** *No five distinct points on a circle embed as box vertices or as vertices of a product of regular simplices. Every such set has strictly positive sup distance of its squared-edge data from* CUT_5.

Proof. If they were box vertices, every four-point subset would be a rectangle. Fix three vertices of one rectangle. They are noncollinear and form a right triangle. Its longest side identifies the hypotenuse and hence the right-angle vertex; the remaining rectangle vertex is uniquely the sum of the hypotenuse endpoints minus that vertex. The fifth point, together with the fixed three, would be another rectangle with the same completion and would equal the fourth point, a contradiction.

It follows that some four-point subset is nonrectangular. Its squared edge vector is outside CUT_4, so its exact margin $\eta(q|_I) > 0$ follows from the four-point theorem. Restriction of cut vectors from five labels to I maps CUT_5 into CUT_4 and is contractive in sup norm. Therefore

$$\mathrm{dist}_\infty(q, \mathrm{CUT}_5) \ge \max_{|I|=4} \eta(q|_I) > 0.$$

No Ramsey obstruction has been inferred. The five-point positive class from the source necessarily uses a mechanism beyond these vertex hosts. □

Example 9.6 (Five-point exact primal and dual certificate). The following certificate was checked by exact rational arithmetic [20]. A numerical linear program was used only for discovery; the proof below uses its explicit primal and dual certificates. Use labels

$$(1,0),\ (0,1),\ (-1,0),\ (0,-1),\ (3/5,4/5).$$

In lexicographic edge order $(12,13,14,15,23,24,25,34,35,45)$,

$$q = (30,60,30,12,30,60,6,30,48,54)/15,$$
$$d = (26,56,34,16,34,56,10,26,44,50)/15.$$

The seven positive cut weights for d are

$\{3\}$	$\{2,3\}$	$\{4\}$	$\{1,4\}$	$\{3,4\}$	$\{1,3,4\}$	$\{2,3,4\}$
$2/15$	$8/15$	$2/15$	$14/15$	2	$2/15$	$2/15$

For example edge 35 crosses all cuts containing 3 in the table, yielding $(2+8+30+2+2)/15 = 44/15$. Direct cut evaluation gives all ten displayed entries, and $\|q-d\|_\infty = 4/15$. Conversely $q_{35} - q_{23} - q_{25} = 4/5$; its triangle dual witness and (9.3) give the lower bound $4/15$. Hence

$$\boxed{\operatorname{dist}_\infty(q, \mathrm{CUT}_5) = 4/15.}$$

This equality is established by a cut-cone primal and a dual witness, not by a false identity $\mathrm{CUT}_5 = \mathrm{MET}_5$. Again $P = \mathrm{diag}(-1,1,1)$ is a rational tensor certificate for the original five points. The repaired box need not be coplanar, and its ordinary distances are not those of the original configuration.

10 Ram formulation: gains, costs, and nonimplications

The useful reformulation retains two different data systems. The affine tensor system retains exact coordinate-field relations and the physical gradient Gram. The cut system retains nonnegative host weights, exact squared distances, and their separating functionals. They answer different questions and should not share a success flag.

For a finite rational basis, a change of coordinates $w' = wS$, $S \in \mathrm{GL}_k(\mathbb{Q})$, changes U to US and C to $S^{-1}CS^{-T}$, leaving UCU^T and all tensor evaluations unchanged. This identity follows by substitution in the tensor expansion. It is an exact representation choice, not a stronger classification theorem. Likewise, for a real invertible edge-data map R, a cut vector $q = A_{\mathrm{cut}}w$ becomes $Rq = (RA_{\mathrm{cut}})w$, still with $w \geq 0$. The correct coefficient norm is $\|R^{-1}z\|_\infty$, and a dual witness becomes $R^{-T}y$. Indeed $(R^{-T}y)^T Rq = y^T q$. Dropping the transported cone or metric invalidates the certificate. This is the applicable finite-cone lesson of the actual theorem `thm:transport` in [19].

Advantages. The signed stencil makes the full evaluation histogram visible. Proposition 3.2 isolates a genuine Taylor remainder, and Proposition 4.3 gives a finite budget for rational coefficient replacement without corrupting exact constraints. The number-field lift produces explicit certificates beyond a chosen quadratic interpolation test. Cut weights and dual margins prevent wasted searches for impossible box encodings. None of these advantages requires an unsupported assertion of universal transform invertibility.

Costs and limitations. Rational relation discovery for arbitrary real inputs is not supplied. The lattice ideal intersection, smoothing support, denominator clearing, permutation orbits, and Hales–Jewett length may all be very large. The nonconstructive necessity coloring and the ultrafilter proof are valid for arbitrary colorings but give no practical coloring algorithm or useful upper bound on $D(A, r)$. A finite positive-semidefinite moment relaxation loses signed cancellation. An approximate zero in the tensor system is not an exact zero. A finite transitive-host failure is not a Ramsey failure, and the five-point box obstruction does not certify non-subtransitivity. The nine-point density corollary shows why noisy geometry alone cannot resolve the field-sensitive classification.

Future work. First, replay the full pinned Lean dependency chain and compare its definitions to the manuscript, not just its theorem name. Second, implement exact number-field certificates with explicit multiplication tables and extraction of P. Third, extract finite stencil support and quantitative rationalization bounds for small non-box configurations. Fourth, make endpoint costs and the forcing dimension effective without changing the order of color and scale choices. Fifth, examine six-point circle configurations using full tensor relations, rather than treating rank-five interpolation failure as a negative result. Finally, separate geometric measurement certification from the field-theoretic input contract in any software interface.

11 Audit register, public refresh, and formalization evidence

11.1 Step-level status

In this register, "checked" means a direct reading and mathematical rederivation above, not proof-assistant replay. No displayed source contradiction was found in this audit. The source's main theorem should not be described as established merely because its catalogue level is L5. That level records direct applicability to the private Ramsey problem.

Source step	What was checked and what is not claimed
`alg:equivalence`, `alg:noetherian`, `alg:congruence`	Rational versus real scalar fields, the free B-module expansion, descent in the full matrix space, flip symmetry, localization, and congruence transport were checked. No numerical field-recognition algorithm is inferred.
`alg:necessity`	Quotient separation, extension of congruent copies by a linear isometry, the -1 invariant, and finite modular coloring were checked. The local bound drops a redundant functional. Algebraic choice is used; the avoiding coloring was not enumerated.
`stencil:finite`	Coset kernels, formal log/exp, the quadratic ideal condition, flatness preserving a finite intersection, and finite truncation inside the ideal were checked. Completion flatness is an external theorem, whose exact Noetherian hypothesis was refreshed.
`pairs:bump`, `pairs:rational`, `pairs:two-scale`	Schur positivity, Gaussian differentiation for indefinite C, weighted cutoff error, Riemann sums, rational approximation in a fixed convolution image, and exact PSD completion were checked. No explicit small-η smoothing function or general stencil extractor was run.
`groups:pairs`, `groups:fractions`	Fixed-distance connectivity, inverse factors, normality, the two supported multiplication rules, and derived-depth induction were checked. Arbitrary MM^{-1} closure is not assumed.
`groups:averaging`, `groups:endpoints`	Right normalization, derived quotient averaging, the $(h-1)/h$ relative-cost contraction, preservation of other orbit bounds, and the correction ratio were checked. No effective global degree bound or exhaustive free-group computer verification was obtained.

Source step	What was checked and what is not claimed
`paths:synchronization`, `paths:corner-group`, `paths:realization`, `paths:criterion`	All target weights are fixed before the word coloring; common strings align; the corner identity is p; the final multiplier is exactly $t/t = 1$. The compact finite-dimension reduction was checked. Finite Hales–Jewett and ZFC compactness are declared inputs, not finite experimental conclusions.
`cons:quadratic`, `cons:quadrilateral`, `cons:transitive`, `cons:kites`	The adjugate lift does not invert a tensor determinant. Circle interpolation and transitive averaging were checked. The kite's non-subtransitivity uses the external LRW theorem, not our box test.
`cons:nine`, `cons:twelve`	The domain localization, legitimate determinant specialization, and all derivative moment cancellations were checked. The determinant and rational specializations were tested exactly. No finite numerical test certifies algebraic independence or Liouville transcendence; those properties enter through the stated proofs.

11.2 Public primary literature refreshed on October 7, 2026

Only public titles, identifiers, and public source URLs were sent to web tools. No private corpus text was uploaded. Leader–Russell–Walters' primary paper [4], Theorem 1 and Corollary 2, confirms the non-subtransitive transcendental kite; its argument and statement were read, with the theorem retained as an external input. The finite transitive-set conjecture is recorded in [3]. The Stacks Project's Tag 00MB states flatness for completion of a Noetherian ring at an ideal; this is precisely the hypothesis needed in Section 3.

Pálvölgyi's September 20 preprint [15] presents a seven-point cyclic non-Ramsey construction. Its main proof and the relevant appendix statements were read; its author explicitly labels the appendix claims as unchecked. Theorem 7.7 supplies an independent proof of the particular algebraic-independence claim in its Theorem A.3 through the tensor bridge. The rest of the appendix has not been validated here. The source's nine-point determinant argument does not depend on it. The public author's note also announces a transitive-set proof under exposition; an announcement is not imported as an independently checked theorem here. Consequently the private report's September 7 openness language must be date-labeled, not repeated as a current assessment.

The refreshed primary abstracts of Ivan–Leader–Walters [10] record the block-size obstruction and their special 123 template result. Their generalized-prism paper [11] preserves soluble symmetry in its Ramsey route; it is not used to claim all transitive sets without the new sufficiency argument. Behague's v3 [12] studies the subsoluble status of known classes. Moore [13] and Mirabi [14] state one-point extension theorems with the new point *outside the affine hull*. These are not a proof for arbitrary additional points on the same circle. These recent papers were refreshed at abstract and version-history depth, not re-audited proof by proof. The historical foundational references [2, 5, 6, 8, 9] identify the classical inputs; their complete published proofs were not all reread.

11.3 Existing Lean formalization: neither ignored nor overcredited

The pinned public `lean/formalization.yaml` lists `OAI.EuclideanRamsey.classification`, its solution file, and comparator configuration [16]. The local files listed in Section 12 were all read here. The challenge defines Ramsey-ness with $r \geq 2$, a finite $D \geq 1$, arbitrary maps to $\mathrm{Fin}\, r$, and exact pairwise distances. Its `FieldCriterion` is over the coordinate field tensor product over $\mathbb{Q}$, with the augmented evaluation zeros and multiplied identity spatial block. The declaration requires $2 \leq s$, $1 \leq d$, injectivity, and full affine span, matching the paper.

The solution `Main.lean` imports `CorrectionPaths`, gives an explicit proof using the tensor/two-scale/path chain, and also proves `full_affine_representative` and `classification_nonempty`. The comparator permits `propext`, `Quot.sound`, and `Classical.choice`, and has `enable_nanoda: false`. The challenge's `sorry` is a specification placeholder, not evidence of a gap in the distinct solution module. Neither Lean, its full import chain, nor the comparator was replayed in this task. Thus this is positive, bounded formalization evidence, not a kernel-verification claim.

11.4 Executed exact checks and outstanding gates

The local checker uses only Python's standard-library `fractions.Fraction`. Its 217 checks pass. They include all ten source-build hashes; the 9×9 determinant specialization; circle evaluation ranks for one through nine rational parameters; the $\mathbb{Q}(\sqrt{2})$ idempotent and lifted sphere identities; a signed two-label stencil and its convolution Hessian; a rounding-budget instance; 24 finite synchronization lists; the four-point cut nullspace and 128 exact cut-weight reconstruction instances; the line, kite, and local five-point certificates; and formal square-moment identities at four rational parameter values.

A separate narrow quantifier audit was read in full, together with its test script, and the script was rerun. It reports no concrete defect in the examined arguments of source Sections 1–6, with particular attention to field descent, rational-linear separation, exact stencils, indefinite C, synchronization, and finite-dimensional compactness. Its independent finite checks comprise 120 rational synchronization cases and 2,460 target weights, plus the two-point stencil, the tensor-multiplication and nonclosed-projection witnesses, and modular-color and final-ratio margin checks; all passed. These counts are separate from the 217-check suite. The weaker-replacement counterexamples have been incorporated above as warnings, not represented as source defects. This audit did not independently validate Section 7, replay formal imports, or reprove the external classical inputs.

The separate seven-point certificate script and its generated evidence were also read in full and replayed. All 141 exact assertions passed: three rational node sets check Lagrange moments, $S = 0$, symmetry and constant-block cancellation for M, energy 4, and the orientation of the pure-tensor product rule with independently assigned coefficient jets. These are substitutions in rational identities, not derivations of $\mathbb{Q}$ or non-Ramsey examples with rational coordinates. The general proof is Proposition 7.6 and Theorem 7.7, not an inference from those samples.

The checker is deliberately not a theorem-proving surrogate. It does not construct the all-dimensional avoiding coloring, test every coloring, replay Hales–Jewett, validate a generic ideal computation, or decide algebraic independence of real inputs. In particular the classification's positive conclusions retain the full sufficiency dependency, while the negative seven-, nine-, and twelve-point conclusions use necessity only. Formal import replay and independent novelty evaluation remain open validation tasks. No claim about another batch manuscript is made here.

12 Reading and reproducibility ledger

The paths below identify the exact local mathematical sources and reproducibility evidence. Reading a discovery map or a theorem declaration is distinguished from reading a proof or replaying a formal development.

Pinned manuscript. Full reading of the ten build files, with every hash matching the local source inventory. The root is

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/A-classification-of-finite-Euclidean-Ramsey-configurations-September-23-2026/build

main.tex, refs.bib, figures/synchronization.tex;
01-introduction.tex, 02-tensors.tex, 03-stencil.tex,
04-pairs.tex, 05-paths.tex, 06-groups.tex, 07-consequences.tex.

All seven numbered files reside in the sections/ subdirectory. Catalogue row 318, source inventory row 318, snapshot.json, and the relevant source-hashes.json entries were inspected.

Private direct source. Full file, all proofs, appended refinements, limitations, and bibliography:

C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-229-astra.tex

The exact directory name contains spaces. Particularly used: thm:cut, thm:four, thm:rectangle, prop:robust, thm:weights229, thm:repair229, and eq:dualmargin229. Its inherited 4,097-check suite was *not* rerun; the present 217-check suite is separately scoped.

Corpus discovery and actual theorem reads. Targeted discovery searches were made in all three local maps corpus-core-map.txt, corpus-foundations-map.txt, and corpus-applications-map.txt under C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit. The maps are routing evidence, not substitute theorem proofs. The actual foundation read was

C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex

The first 300 lines were read, including thm:ram-local, its integral and absolute-moment remainder, signed/second-order conventions, source-centered qualifications, and thm:degree-closure. Included companion chapter files and the whole book were not read. The application-theorem read was

C:/Users/mural/OneDrive/Documents/RamMathUnsolved/TR-meta1-math.tex

The selected passage on exact quadratic measurements, finite coordinate transport, thm:transport, redundant banks, and the deterministic correction propositions was read directly. Only the elementary exact transport theorem is used here; no probabilistic ellipsoid theorem is imported. No assertion is made that all reports mentioned in the maps were read.

Formalization and local certificates. Full local reads of

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/lean-batch5/OAI/Combinatorics/EuclideanRamsey/Main.lean
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/lean-batch5/ComparatorChallenges/EuclideanRamsey.lean
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/lean-batch5/ComparatorChallenges/EuclideanRamsey.json

Their imported proof dependencies were not read or executed here. The five-point script and its JSON certificate were read at

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M318_parent_five.py
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M318_parent_five.json

The original LP was not rerun. Its positive cut weights and triangle witness were independently replayed with exact fractions in our checker.

The independent quantifier audit and complete test script were read at

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M318-independent-quantifier-audit.txt
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M318-independent-quantifier-tests.py

The latter was rerun with exact rational arithmetic. Its mathematical scope and limits are recorded above.

The seven-point derivation review, exact test script, and supplied JSON were read in full at

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M318-parent-seven-point-review.txt
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M318_seven.py
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M318_seven.json

The script was replayed with its generated output directed to the local arithmetic-evidence directory. The public proof and coefficient choice in [15, Appendix, Theorem A.3] were also read directly; only this scoped appendix claim is independently established here.

Exact arithmetic evidence. The report's checker and its JSON results are in

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M318-Euclidean-Ramsey-Configurations/M318-author-evidence/M318_exact_checker_20261007.py
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M318-Euclidean-Ramsey-Configurations/M318-author-evidence/M318_exact_checker_20261007.json

The same directory contains `M318_seven_replay_20261007.py` and its JSON output, preserving the original supplied test and certificate. These finite identities complement, and do not replace, the proofs and the explicitly identified external inputs.

13 Conclusion

The source supplies a coherent field-sensitive route from exact tensor annihilation to arbitrary-color forcing. Its delicate points are the flat ideal-intersection lift, approximation restricted to a fixed exact convolution image, supported noncommutative fractions, and preparing all Hales–Jewett variable-count scales before word coloring. These steps have been read and reconstructed rather than replaced by a generic Ram transform slogan.

The rigorous local additions are exact algebraic lifts, finite-field-data certificate alternatives, a preserved-constraint rounding budget, a smaller obstruction-color bound, a derivation-to-tensor bridge with a seven-point application, and explicit separation of the five-point positive class from box embeddings. The cut cone remains a host criterion, not the Ramsey criterion. The formalization already present at the pinned source is acknowledged at its actual inspected depth. Finite arithmetic, mathematical proof reconstruction, imported classical results, and unreplayed formal evidence remain separate levels of support.

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